

Homework 3 solution:

Q1. Convolution Neural Networks

- A) Link as a reference: <https://towardsdatascience.com/understanding-and-calculating-the-number-of-parameters-in-convolution-neural-networks-cnns-fc88790d530d>

Layer	Dimensions of Activation map	# of parameters
INPUT	512 * 512 * 3	0
CONVOLUTION(7, 16)	254 * 254 * 16	2368
POOLING(2)	127 * 127 * 16	0
CONVOLUTION(5, 64)	63 * 63 * 64	25664
POOLING(3)	21 * 21 * 64	0
CONVOLUTION(3, 64)	11 * 11 * 64	36928
POOLING(2)	5 * 5 * 64	0
FULLY-CONNECTED(5)	5 * 1	8005

- B) i) Convolution :

Convolution on Red channel :

$$\begin{bmatrix} 1 & 3 & 4 & 2 & 5 & 0 \\ 3 & 5 & 2 & 1 & 2 & 1 \\ 2 & 1 & 4 & 3 & 3 & 3 \\ 1 & 2 & 3 & 2 & 5 & 2 \\ 3 & 1 & 5 & 4 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 3 & 1 & 3 \\ 1 & 2 & -1 \end{bmatrix}$$
$$\begin{bmatrix} 17 & 19 \\ 12 & 37 \end{bmatrix}$$

Convolution on Green channel :-

$$\begin{bmatrix} 1 & 3 & 4 & 4 & 5 & 0 \\ 3 & 5 & 2 & 2 & 2 & 1 \\ 2 & 1 & 4 & 6 & 3 & 3 \\ 1 & 2 & 3 & 4 & 5 & 2 \\ 3 & 1 & 5 & 8 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 3 & 1 & 3 \\ 1 & 2 & -1 \end{bmatrix}$$
$$\begin{bmatrix} 17 & 26 \\ 12 & 47 \end{bmatrix}$$

Convolution on Blue Channel:-

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 0 \\ 3 & 5 & 2 & 1 & 2 & 1 \\ 2 & 1 & 4 & 3 & 3 & 3 \\ 1 & 2 & 3 & 2 & 5 & 2 \\ 3 & 1 & 5 & 4 & 3 & 0 \end{pmatrix} \quad \begin{bmatrix} 1 & 0 & -1 \\ 3 & 1 & 3 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 17 & 19 \\ 12 & 37 \end{bmatrix}$$

Convolution on all channels:-

$$\begin{bmatrix} 51 & 64 \\ 36 & 121 \end{bmatrix}$$

ii) Average Pooling:

padding = 1
Stride = 1

padding

0	0	0	0
0	51	64	0
0	36	121	0
0	0	0	0

$$\begin{bmatrix} 0 & 0 & 0 \\ 25 & 57 & 32 \\ 18 & 78 & 60 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 57 & 32 \\ 18 & 78 & 60 \end{bmatrix} \quad \text{Average pooling}$$